

Software Requirements Specification (SRS)

Project: Leave Management System (LMS)

Version: 1.0

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1. Introduction

1.1 Purpose

The Leave Management System (LMS) is designed to simplify and automate the process of managing employee leaves within an organization. By providing a streamlined interface, automated workflows, and real-time updates, we aim to eliminate manual paperwork, reduce administrative overhead, and ensure that leave records are always up to date. The system will allow employees to easily apply for leave, managers to review and approve requests promptly, and HR personnel to track leave balances and generate insightful reports.

This SRS details the functional and non-functional requirements for the initial release of the Leave Management System, version 1.0.

1.2 Scope

The Leave Management System (LMS) is designed to streamline and automate the entire leave management process within the organization. The system primarily covers:

- **Employee Leave Requests:** Employees can apply for different types of leave (casual, sick, earned, etc.) through the system, view their leave balances, and track the status of their applications.
- **Manager Approvals:** Managers can review, approve, or reject leave requests based on organizational policies and staffing requirements. The system ensures that approval workflows are handled digitally without manual intervention.
- **HR Monitoring:** The HR department has access to oversee all leave applications, generate compliance checks, and maintain centralized records of leave histories. HR can also set policies like carry-forward rules, leave quotas, and encashment limits.
- **Reporting:** The system provides detailed reports on employee leave records, loss of pay, leaves encashed.

1.3 Audience

This document is intended for the following audiences:

Direct Audience (who actively use the SRS):

- **Developers** – To design and implement the system as per the requirements.
- **QA Engineers** – To derive test cases and verify functional and non-functional requirements.

- **Project Managers** – To monitor scope, allocate resources, and track project progress.
- **Business Analysts** – To ensure requirements align with business goals and resolve ambiguities.
- **Documentation Writers** – To prepare user manuals, FAQs, and training material.
- **IT System Administrators** – To manage deployment, security, integration, and maintenance.

Indirect Stakeholders (whose needs or policies are captured in the SRS but who may not directly use this document):

- **HR Managers** – To validate that organizational leave policies (types, quotas, workflows, compliance rules) are correctly documented.
- **Employees** – End users of the system; their requirements and usability needs are represented in this document.

1.4 Definitions

LMS - Leave Management System

HR - Human Resources

UI - User Interface

API - Application Programming Interface

RBAC - Role-Based Access Control

2. Overall Description

2.1 Product Perspective

The LMS is a web-based HR tool that integrates with the organization's employee database and works in alignment with payroll and attendance systems. It serves as a module within the broader HR management ecosystem, ensuring that leave policies are enforced consistently. The system will rely on existing employee records and provide data for downstream processes such as payroll calculations and compliance reporting.

2.2 Major Product Functions

- User authentication
- Manager approval/rejection workflow
- Leave balance tracking
- Calendar View
- Role-Based Dashboards
- Notifications via email
- Reports

- Policy configuration by HR
- Audit logs

2.3 User Roles

- Employee:
 - Apply for different types of leave (casual, sick, earned, etc.).
 - Edit or delete leave requests that are still pending approval.
 - Check current leave balance in real time.
 - View leave history and track approval status.
- Manager:
 - Review leave applications submitted by employees.
 - Approve or reject leave requests based on team requirements and policies.
 - Generate reports
- HR/Admin:
 - Set up initial leave rules (types, quotas, carry-forward limits, encashment policies).
 - Update policies in the system whenever company policies or labor laws change.
 - Generate reports

2.4 Operating Environment

The Leave Management System (LMS) will operate as a **web-based application** accessible through standard web browsers (such as Google Chrome, Microsoft Edge, etc).

The system will be hosted on a **central server** with a relational database to store employee records, leave transactions, and audit logs.

An **email notification service** will be integrated to send alerts and updates, including leave application submissions, approvals and rejections.

2.5 Constraints

- **Role-Based Access Control (RBAC):** Access levels must be enforced for employees, managers, and admin users.
- **Secure Authentication:** User login credentials must be encrypted and securely stored (e.g., password hashing).
- **Data Privacy:** The system must ensure that only authorized users can access leave-related data.
- **Scalability:** The system should handle an increasing number of employees without performance degradation.

- **Availability:** The LMS should remain accessible during working hours with minimal downtime.

3. External Interface Requirements

3.1 User Interfaces

- **Primary UI:** A web-based dashboard designed for easy navigation. The interface will provide menu options for applying leave, approving requests, and tracking status.
- **Role-Based Views:**
 - **Employee:** Apply for different types of leave, check leave balance, view request history, and track current application status.
 - **Manager:** Review and approve/reject leave requests from team members and view team availability and leave reports.
 - **HR:** Configure leave types and quotas, and view pending leave reports.
- **Accessibility:** The UI will be simple, user-friendly, and designed to support clear navigation for all users.
- **Notifications:** The system will provide email notifications to inform employees and managers about leave status updates.
- **Self-Service Design:** Employees will have access to a unified dashboard where they can check leave balances, review their leave history, and track pending approvals without needing HR/admin assistance.

3.2 Hardware Interfaces

- **Desktop/Laptop:** The system will be accessible from standard desktops or laptops used by employees, managers, and HR staff.

3.3 Software Interfaces

- **Database Interface:** The system will interact with a central relational database (e.g., MySQL/PostgreSQL) to store employee details, leave applications, approvals, balances, and reports.

- **Notification Service:** The system will use an email service to send notifications for leave requests, approvals, rejections, and reminders.
- **Reporting Interface:** The system will generate downloadable reports (e.g., PDF, Excel) for HR on leave history, loss of pay, and encashments.

3.4 Communications

- **Secure Access:** All interactions will happen through web browsers using HTTPS (if hosted on a secure server).
- **Role-Based Access Control:** Employees, Managers, and HRs will access different modules based on their login role.
- **Session Management:** The system will maintain secure sessions to prevent unauthorized access.

4. System Features (Functional Requirements)

At least 15 functional requirements are defined, grouped by feature:

4.1 User Management

Description: Manages employee data and roles.

Req ID	Requirement	Type	Priority	Source/Stake holder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-001	The system shall allow assignment of predefined roles (Employee, Manager, HR).	Functional	High	HR	When HR assigns a role (Employee, Manager, Admin), the role is saved and permissions apply correctly.	Depends on role-based access control (RBAC) module.

LMS-F-002	The system shall validate required fields before saving; if missing, display: "Please fill in all required fields".	Functional	Medium	Employee / HR / Manager	When an employee leaves a required field blank and saves, the system shows "Please fill in all required fields."	Depends on form validation rules.
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4.2 Leave Application and Approval

Description: Allows employees to apply for leave and managers to approve or reject requests.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-011	The system shall allow employees to submit leave requests with start date, end date, leave type, and reason.	Functional	High	Employees	When an employee submits a leave request with all details, the system accepts and processes it.	Needs leave type definitions.
LMS-F-012	The system shall reject requests with start dates in the past(before	Functional	High	Employees	When an employee submits a leave request with a past start date(before the last 15 days), the system shows	Requires date validation logic.

	the last 15 days), showing: "Invalid leave start date".				"Invalid leave start date."	
LMS-F-013	The system shall check available leave balance before submission; if exceeded, show: "Insufficient leave balance".	Functional	High	Employees	When an employee submits a leave request exceeding balance, the system shows "Insufficient leave balance."	Depends on Leave Balance View .
LMS-F-014	The system shall route leave requests to the appropriate manager.	Functional	High	Managers, Employees	When an employee submits a leave request, it is routed to the correct manager.	Needs org hierarchy mapping.
LMS-F-015	The system shall allow managers to approve, reject leaves.	Functional	High	Managers	When a manager reviews a request, they can approve, reject, or request modification successfully.	Requires workflow rules.

4.3 Leave Balance View

Description: Tracks leave usage and displays real-time balances.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-021	The system shall maintain a leave ledger for each employee.	Functional	High	Employees, HR	When an employee checks leave balance, the system displays their leave ledger.	Needs DB structure.
LMS-F-022	The system shall update leave balances after approval.	Functional	High	Employees, Managers	When a leave is approved, the leave balance updates correctly.	Depends on leave approval workflow.
LMS-F-023	The system shall support partial-day leave tracking.	Functional	Medium	Employees	When an employee applies for partial-day leave, the balance reflects the partial deduction.	Needs fractional leave calculation.

LMS-F-024	If data retrieval fails, show: "Unable to retrieve leave history. Please contact the administrator".	Functional	Medium	Employees	When leave history cannot be retrieved, the system shows "Unable to retrieve leave history. Please contact the administrator."	Error handling dependency.
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4.4 Types of Leaves

Description: Defines different leave categories and policy rules.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-031	The system shall allow defining and editing leave types (e.g., Sick, Casual, Earned).	Functional	High	HR	When HR creates or edits a leave type, it is available for employees to use.	Needed before leave application works.
LMS-F-032	The system shall allow configuration of accrual rules per leave type .	Functional	Medium	HR	When accrual rules are configured, balances update according to those rules.	Accrual format .

LMS-F-033	The system shall apply carry-forward (30 days max) for earned leaves.	Functional	Medium	HR	When carry-forward or encashment policies are applied, balances update accordingly.	Depends on company leave policy.
LMS-F-034	The system shall integrate a holiday calendar to adjust calculations.	Functional	Medium	HR, Manager, Employees	When a leave overlaps with holidays, the calculation excludes holiday days.	Needs calendar integration.

4.5 Cancel Leave

Description: Allows employees to cancel pending leave requests before supervisor action.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-041	The system shall allow employees to cancel pending leave requests before supervisor action.	Functional	Medium	Employees	When an employee cancels a pending leave request, the system updates the request.	Depends on leave workflow.

LMS-F-042	The system shall allow employees to submit leave cancellation requests for approved leaves.	Functional	High	Employees	When an employee submits a leave cancellation request, the system fetches details of approved leave and processes it.	Needs status validation.
LMS-F-043	The system shall route leave cancellation requests to the appropriate manager.	Functional	High	Managers, Employees	When an employee submits a leave cancellation request, it is routed to the correct manager.	Needs org hierarchy mapping.
LMS-F-044	The system shall allow managers to approve, reject leaves cancellation.	Functional	High	Managers	When a manager reviews a cancellation request, they can approve or reject successfully.	Requires workflow rules.

4.6 User Access

Description: The system shall provide secure user access through login and logout functionality. Each employee, manager and HR will have unique credentials.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-051	The system shall allow users to log in with unique credentials (username and password)	Functional	High	Employee, HR, Manager	When a user enters a valid username and password, the system grants access to their account.	Depends on credential storage and authentication service.
LMS-F-052	The system shall authenticate credentials against stored user data.	Functional	High	Employee	When login is attempted, the system validates credentials against securely stored data.	Requires password hashing & encryption.
LMS-F-053	The system shall grant access based on role (employee, manager)	Functional	High	Employee, HR, Manager	When a valid user logs in, only role-specific permissions are granted.	Depends on RBAC configuration.

LMS-F-054	The system shall deny access for invalid credentials and display an error message.	Functional	High	Employee, HR, Manager	When invalid credentials are entered, the system denies access and shows a generic error.	Must not reveal whether the username or password was incorrect.
LMS-F-055	The system shall allow users to log out at any time.	Functional	High	Employee, HR, Manager	When logout is clicked, the session ends and the login page is shown.	Depends on session management module and authentication framework.
LMS-F-056	The system shall log all login and logout activities for audit purposes	Functional	High	Auditing	When login/logout occurs, the system records user ID, timestamp, and action.	Depends on centralized logging service or database table for audit trails.

4.7 Error Handling

Description: Detection, logging, and displaying of clear error messages.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
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LMS-F-061	The system shall display meaningful error messages for invalid inputs (e.g., wrong login credentials).	Functional	Medium	Employee, HR, Manager	When invalid data (e.g., wrong password) is entered, the system shows a clear message.	Must not leak technical info.
LMS-F-062	The system shall guide users with corrective actions when possible.	Functional	Medium	Employee, HR, Manager	When an error occurs, the system suggests corrective steps (e.g., "Re-enter password").	Improves usability.

4.8 Notifications and Alerts

Description: Sends notifications and alerts to employees and supervisors for important events such as leave application submission, approval, rejection, cancellation, and policy updates.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-071	The system shall send notifications to employees when their leave is approved, rejected, or	Functional	High	Employee	When leave status changes, the employee receives a notification.	Depends on notification channel (email, app, SMS).

	canceled.					
LMS-F-072	The system shall notify managers when a new leave or leave cancellation request is submitted.	Functional	High	Manager	When an employee submits leave, the manager is notified.	Depends on the workflow engine.
LMS-F-073	The system shall ensure notifications are delivered in real time.	Non-Functional	High	Employee, Manager	When events occur, notifications are delivered instantly.	Depends on messaging service.

4.9 Reports and Analytics

Description: Generates reports and analytics to provide insights into employee leave usage, balances, and trends.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LMS-F-081	The system shall allow HR and managers to generate leave balance and usage reports.	Functional	High	HR / Manager	When requested, the system generates reports for leave balances and usage.	Depends on the reporting engine.

LMS-F-08 2	The system shall support filters like employee role.	Functional	Medium	HR, Manager	When filters are applied, reports update accordingly.	Requires indexed / queryable data.
LMS-F-08 3	The system shall allow exporting reports to PDF or Excel formats.	Functional	Medium	HR, Manager	When exporting, reports are generated in PDF/Excel formats.	Depends on export libraries.
LMS-F-08 4	The system shall handle errors gracefully if the requested report contains no data (e.g., "No records found").	Functional	Medium	HR, Manager	When no records exist, the system displays "No records found."	Prevents confusion/error screens.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs LMS-NF-###.

Req ID	Requirement	C ategory	Priority	Acceptance criteria / Measurement
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LMS-NF-001	The system shall provide an intuitive UI so that employees, managers, and HR can perform core tasks (apply leave, approve/reject, generate reports) without extra training.	Usability	High	Users can complete core tasks (apply, approve/reject, generate reports) successfully without needing extra help
LMS-NF-002	The system shall accurately calculate leave balances and remain stable under valid/invalid inputs, concurrent requests, and server restarts.	Reliability & Stability	High	Leave balances update correctly, system does not crash or lose data even with invalid inputs, many users at once, or after a restart.
LMS-NF-003	The system shall support at least 500 employee records and 20 concurrent users without performance degradation.	Scalability	Medium	The system runs smoothly with 500 employees in the database and at least 20 users applying for leave at the same time.

5.1. Security

5.1.1 Security Objectives

1. Ensure confidentiality of employee and organizational data.
2. Ensure integrity and accountability of leave transactions.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
PRJ-SR-001	The system shall include automated daily backups to prevent loss of leave data in case of hardware or software failure.	Security	High	- Backups run automatically every 24 hours. - Backup files are stored securely in database.
PRJ-SR-002	The system shall require secure login with a unique username and password for all users.	Security	High	- User must log in with valid username/password before accessing the system. - Password will be encrypted and stored.
PRJ-SR-003	The system shall implement role-based access control (RBAC) to ensure that employees, managers, and administrators access only the features and data relevant to their roles.	Security	High	- Employees can view their leave balance and submit requests but cannot see others' data. - Managers can approve/reject requests for their team only. - HR have full access including audit trails and configurations. - Unauthorized access attempts are logged.

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

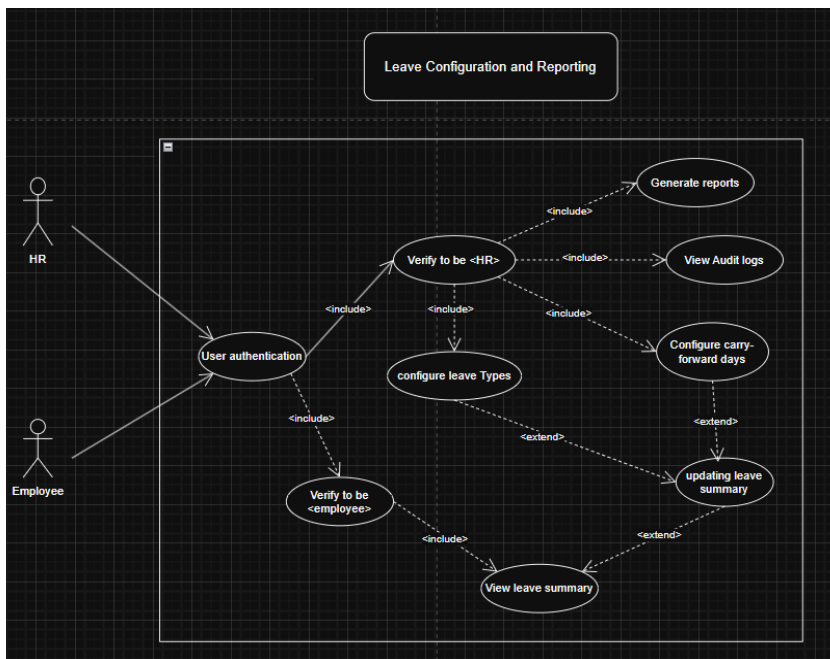
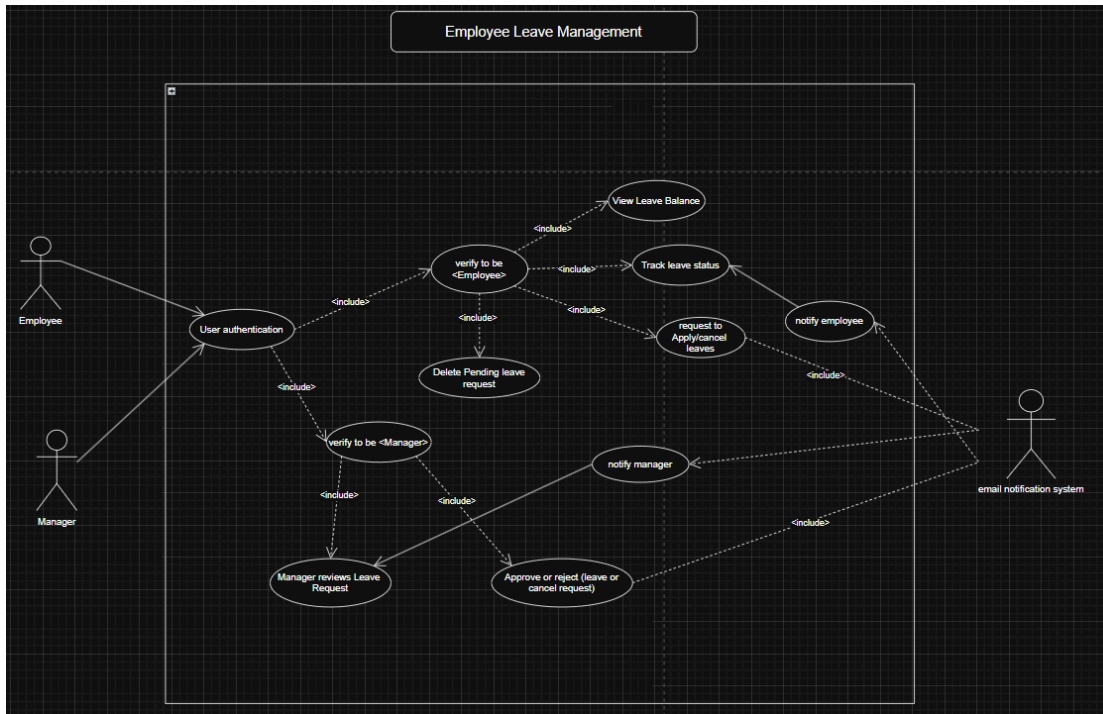
- **Usability:**
The system shall provide a simple and intuitive user interface. Employees, managers, and HR staff should be able to perform tasks (apply leave, approve/reject, generate reports) with minimal training.
- **Reliability & Stability:**
The system shall perform leave calculations accurately and remain stable under valid and invalid inputs, concurrent requests, and server restarts without data loss.
- **Availability:**
The system shall ensure at least **99.5% uptime during working hours**, excluding scheduled maintenance.
- **Maintainability:**
The system shall be designed with a modular architecture so that bug fixes, updates, and enhancements can be made with minimal impact on other modules.
- **Scalability:**
The system shall support growth in users and transactions.
 - Current version: at least **500 employee records** and **20 concurrent users** without performance issues.
 - Future versions: scalability can be extended with better server configurations.
- **Security:**
The system shall enforce role-based access control, secure session management, and HTTPS communication to protect user data.
- **Auditability:**
The system shall maintain audit logs of all critical actions (leave applications, approvals, policy changes) for accountability and compliance.

6.2 Acceptance Tests

- **Usability Test:** Verify that users can submit a leave request, approve/reject requests, and generate reports without additional guidance.
- **Reliability Test:** Apply and approve multiple leave requests; verify leave balances update correctly and no data is lost after server restart.
- **Availability Test:** Keep the system running during a test period and confirm it remains operational with minimal downtime.
- **Maintainability Test:** Modify a leave policy (e.g., change leave quota) and confirm the update reflects correctly without breaking other modules.
- **Scalability Test:** Populate the database with 500 employee records and perform 20 simultaneous leave applications. Confirm the system remains responsive and processes all requests accurately.
- **Security Test:** Attempt to access restricted modules with a lower-privilege role (e.g., employee trying to edit policies). Confirm access is denied.
- **Auditability Test:** Perform leave applications and approvals, then check the audit logs to confirm all actions are recorded with correct timestamps and user details.

7. UML Use-Case Diagram

Use case diagram to include Employee, Manager, and HR roles with leave request, approval, reporting, and policy configuration use cases.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
Functional Requirements (FRs)						
LMS-F-001	Assign predefined roles (Employee/Manager/HR)	4.1 / DS-User-01	User Management	TC-User-01	N	Depends on RBAC
LMS-F-002	Validate required fields before saving	4.1 / DS-User-02	User Management	TC-User-02	N	Needs form validation
LMS-F-011	Submit leave request (date, type, reason)	4.2 / DS-Leave-01	Leave Application	TC-Leave-01	N	Needs leave type definitions
LMS-F-012	Reject requests with past start dates	4.2 / DS-Leave-02	Leave Validation	TC-Leave-02	N	Requires date validation
LMS-F-013	Validate leave balance before submission	4.2 / DS-Leave-03	Leave Balance	TC-Leave-03	N	Depends on balance module

LMS-F-014	Route leave request to appropriate manager	4.2 / DS-Workflow-01	Workflow Engine	TC-Leave-04	N	Needs org hierarchy
LMS-F-015	Manager approves/rejects leave	4.2 / DS-Workflow-02	Approval Workflow	TC-Leave-05	N	Requires workflow rules
LMS-F-021	Maintain leave ledger per employee	4.3 / DS-Balance-01	Leave Balance	TC-Balance-01	N	Needs DB structure
LMS-F-022	Update balances after approval	4.3 / DS-Balance-02	Leave Balance	TC-Balance-02	N	Depends on approval
LMS-F-023	Support partial-day leave	4.3 / DS-Balance-03	Leave Balance	TC-Balance-03	N	Requires fractional leave logic
LMS-F-024	Show error if leave history retrieval fails	4.3 / DS-Balance-04	Leave History	TC-Balance-04	N	Error handling required
LMS-F-031	Define and edit leave types	4.4 / DS-Policy-01	HR Policy Module	TC-Policy-01	N	Needed before requests
LMS-F-032	Configure accrual rules per leave type	4.4 / DS-Policy-02	HR Policy Module	TC-Policy-02	N	Accrual rules TBD
LMS-F-033	Apply carry-forward (30 days max)	4.4 / DS-Policy-03	HR Policy Module	TC-Policy-03	N	Depends on company policy

LMS-F-034	Integrate holiday calendar	4.4 / DS-Policy-04	HR Policy Module	TC-Policy-04	N	Requires calendar data
LMS-F-041	Cancel pending leave requests	4.5 / DS-Cancel-01	Leave Application	TC-Cancel-01	N	Depends on workflow
LMS-F-042	Cancel approved leave requests	4.5 / DS-Cancel-02	Leave Application	TC-Cancel-02	N	Requires status validation
LMS-F-043	Route cancellation to manager	4.5 / DS-Cancel-03	Approval Workflow	TC-Cancel-03	N	Needs org hierarchy
LMS-F-044	Manager approves/ rejects cancellations	4.5 / DS-Cancel-04	Approval Workflow	TC-Cancel-04	N	Requires workflow rules
LMS-F-051	User login with unique credentials	4.6 / DS-Auth-01	Authentication	TC-Auth-01	N	Depends on credential storage
LMS-F-052	Authenticate credentials	4.6 / DS-Auth-02	Authentication	TC-Auth-02	N	Requires hashing & encryption
LMS-F-053	Role-based access	4.6 / DS-Auth-03	RBAC	TC-Auth-03	N	Requires RBAC config
LMS-F-054	Deny access on invalid credentials	4.6 / DS-Auth-04	Authentication	TC-Auth-04	N	Must not leak sensitive info

LMS-F-055	User logout	4.6 / DS-Auth-05	Authentication	TC-Auth-05	N	Requires session mgmt
LMS-F-056	Log login/logout activities	4.6 / DS-Audit-01	Audit Module	TC-Audit-01	N	Needs audit DB table
LMS-F-061	Display meaningful error messages	4.7 / DS-Error-01	Error Handling	TC-Error-01	N	Must avoid technical info
LMS-F-062	Suggest corrective actions for errors	4.7 / DS-Error-02	Error Handling	TC-Error-02	N	Improves usability
LMS-F-071	Notify employee on leave status change	4.8 / DS-Notify-01	Notification Service	TC-Notify-01	N	Requires email/SMS
LMS-F-072	Notify manager on new request/cancellation	4.8 / DS-Notify-02	Notification Service	TC-Notify-02	N	Depends on workflow engine
LMS-F-073	Deliver notifications in real time	4.8 / DS-Notify-03	Notification Service	TC-Notify-03	N	Needs messaging service
LMS-F-081	Generate leave balance & usage reports	4.9 / DS-Report-01	Reporting Module	TC-Report-01	N	Depends on reporting engine

LMS-F-082	Apply filters in reports	4.9 / DS-Report-02	Reporting Module	TC-Report-02	N	Requires queryable data
LMS-F-083	Export reports (PDF/Excel)	4.9 / DS-Report-03	Reporting Module	TC-Report-03	N	Needs export libraries
LMS-F-084	Show “No records found” when report is empty	4.9 / DS-Report-04	Reporting Module	TC-Report-04	N	Error handling

Non-Functional Requirements (NFRs)

LMS-NF-001	Provide intuitive UI (minimal training)	5 / DS-NF-UI	User Interface	TC-NF-UI-01	N	Usability requirement
LMS-NF-002	Accurate leave balance & stable under load	5 / DS-NF-Rel	Core System	TC-NF-Rel-01	N	Reliability requirement
LMS-NF-003	Support 500 employees, 20 concurrent users	5 / DS-NF-Scalability	Core System	TC-NF-Scal-01	N	Scalability
LMS-NF-004	Modular architecture for easy updates	5 / DS-NF-Maint	Core System	TC-NF-Maint-01	N	Maintainability

Security Requirements (SRs)						
PRJ-SR-001	Daily automated backups	5.1 / DS-Security-01	Backup Service	TC-Sec-01	N	Backup & restore
PRJ-SR-002	Prevent deletion of leave records w/o authorization	5.1 / DS-Security-02	Data Protection	TC-Sec-02	N	Only admin can delete
PRJ-SR-003	Secure login (username/password, encryption)	5.1 / DS-Security-03	Authentication	TC-Sec-03	N	Password hashing & encryption
PRJ-SR-004	Role-Based Access Control (RBAC)	5.1 / DS-Security-04	RBAC Module	TC-Sec-04	N	Unauthorized access logged